Comprehensive Comparative Literature Review and Synthesis of Machine Learning and Deep Learning in Financial Fraud Detection

Comparative Analysis of Five Benchmark Studies (2022–2026)

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Abstract

Financial fraud poses an existential threat to economic systems, spanning consumer card transactions, online payment gateways, corporate disclosure falsification, money laundering, and cryptocurrency scams. This document provides a rigorous, multi-dimensional comparative analysis and literature synthesis of five landmark articles in this domain: two breakthrough empirical research papers—**Almazroi & Ayub (2023)** (RXT-J: ResNeXt-GRU with Jaya Optimization) and **Jin & Zhang (2025)** (Dynamic Stacking Ensemble with Feature Importance Weighting)—and three extensive systematic literature reviews and meta-analyses—**Ashtiani & Raahemi (2022)** (Corporate Financial Statement Fraud Detection), **Chen et al. (2026)** (Deep Learning Innovations and the DLSG Framework), and **Hernandez Aros et al. (2024)** (Global Fraud Typology and ML Benchmarking). Five detailed comparative LaTeX tables are provided covering: (1) Bibliographic Metadata and Research Scope, (2) Algorithmic Architecture and Feature Engineering Pipelines, (3) Benchmark Datasets and Imbalance Handling Strategies, (4) Empirical Performance, Evaluation Metrics, and Computational/Latency Trade-offs, and (5) Regulatory Governance (GDPR/CCPA/HIPAA), Explainable AI (XAI), and Future Roadmaps.

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1 Introduction to the Five Landmark Studies

The five examined publications represent the dual technical and survey frontiers of modern artificial intelligence in financial criminology:

[1] Almazroi & Ayub (2023): *"Online Payment Fraud Detection Model Using Machine Learning Techniques"* (*IEEE Access*, Vol. 11, pp. 137188–137203). Proposes an innovative deep learning architecture (**RXT-J**) combining ResNeXt split-cardinality convolutional blocks and Gated Recurrent Units (GRU) fine-tuned using the population-based Jaya optimization algorithm. Feature representations are generated via an Ensemble AutoEncoder and ResNet (**EARN**) ensembler, with SMOTE balancing, PCA, and Isolation Forest anomaly filtering.

[2] Ashtiani & Raahemi (2022): *"Intelligent Fraud Detection in Financial Statements Using Machine Learning and Data Mining: A Systematic Literature Review"* (*IEEE Access*, Vol. 10, pp. 72504–72525). Adopts Kitchenham's systematic review methodology to synthesize 47 peer-reviewed studies (1995–2019) focusing exclusively on corporate Financial Statement Fraud Detection (FSFD). Analyzes data modalities (financial ratios, non-financial governance indicators, textual MD&A, and vocal acoustics) across eight machine learning categories.

[3] Chen et al. (2026): *"Deep learning in financial fraud detection: innovations, challenges, and applications"* (*Data Science and Management*, Vol. 9, art. no. 100162). Synthesizes 108 peer-reviewed publications (2019–2024) and establishes the conceptual **DLSG** (Deep Learning–Sector–Governance) framework, unifying technological innovations (CNN, LSTM, Transformers, GNN, GANs/VAEs) with sector-specific operational constraints and strict regulatory mandates (GDPR right-to-explanation, CCPA, HIPAA, federated learning, and XAI).

[4] Hernandez Aros et al. (2024): *"Financial fraud detection through the application of machine learning techniques: a literature review"* (*Humanities and Social Sciences Communications*, Vol. 11:1130). A comprehensive systematic review and bibliometric analysis (PRISMA and Kitchenham protocols, VOSviewer) examining 104 articles (2012–2023; 86 experimental, 18 surveys). Classifies financial fraud into external (54%) and internal (46%) domains, maps global research clusters, and benchmarks the 10 most prominent algorithms.

[5] Jin & Zhang (2025): *"The analysis of fraud detection in financial market under machine learning"* (*Scientific Reports*, Vol. 15:29959). Designs and validates an advanced Stacking Ensemble learning algorithm integrating heterogeneous base learners (Logistic Regression, Decision Tree, Support Vector Machine, and Neural Network) combined with Recursive Feature Elimination (RFE), feature importance weighting, and dynamic real-time weight adjustment across >1 million transaction records and a 4.2M cross-domain benchmark.

2 Multi-Dimensional Comparative Literature Tables

2.1 Table 1: Bibliographic Metadata, Study Typology, and Research Scope

Table 1: Bibliographic metadata, research nature, targeted fraud domains, and primary objectives of the reviewed studies.

Authors & Citation	Journal & Year	Study Nature	Targeted Fraud Domain	Primary Research Objective / Scope	Key Theoretical Framework / Paradigm
Almazroi & Ayub (2023) [1]	*IEEE Access*, Vol. 11, pp. 137188–137203	Empirical / Original Technical Proposal	Online payment fraud, credit card transactions, electronic financial transfers	To propose a real-time, highly scalable deep architecture (**RXT-J**) capable of capturing complex spatial and temporal transactional representations under extreme class imbalance.	Spatial-temporal hybrid learning (ResNeXt + GRU), multi-stream feature fusion (EARN), and population-based metaheuristic optimization (Jaya).
Ashtiani & Raahemi (2022) [2]	*IEEE Access*, Vol. 10, pp. 72504–72525	Systematic Literature Review (SLR)	Financial Statement Fraud Detection (FSFD), management falsification	To comprehensively examine the ML/DM methods, data sources, ratios, and extraction pipelines employed in 47 FSFD articles (1995–2019) and uncover critical research gaps.	Kitchenham SLR Protocol, Cressey's Fraud Triangle Theory (Incentive, Opportunity, Rationalization), multi-modal feature taxonomies.
Chen et al. (2026) [3]	*Data Science and Management*, Vol. 9, 100162	Systematic Literature Review (SLR)	Cross-sector: Credit cards, banking, insurance, crypto/blockchain, tax	To evaluate advances in Deep Learning models across 108 peer-reviewed publications (2019–2024), mapping technological progress to domain operational realities and regulatory laws.	Kitchenham Framework; introduces the novel **DLSG** (Deep Learning–Sector–Governance) Triadic Meta-Framework.
Hernandez Aros et al. (2024) [4]	*Humanities & Social Sciences Communications*, Vol. 11:1130	SLR and Bibliometric Meta-Analysis	Holistic: Internal (Statements, AML, Tax) vs. External (Credit cards, Loans, Insurance)	To conduct a comprehensive mapping of 104 articles (2012–2023; 86 experimental studies) to identify author networks, geographic dispersion, datasets, models, and evaluation metrics.	PRISMA Guidelines + Kitchenham Methodology; Donald Cressey Fraud Theory; VOSviewer Bibliometric Linkage Mapping.
Jin & Zhang (2025) [5]	*Scientific Reports*, Vol. 15:29959	Empirical / Original Technical Proposal	Financial market transactions, credit card theft, unauthorized transfers, market fraud	To design a robust, high-accuracy Stacking Ensemble framework using dynamic learner weights and feature importance weighting to overcome single-model brittleness and high false alarm rates.	Ensemble Learning Theory (Wolpert's Stacking), Recursive Feature Elimination (RFE), Dynamic Weight Allocation via verification accuracy.

2.2 Table 2: Methodological Architectures, Algorithms, and Feature Pipelines

Table 2: Methodological architectures, learning paradigms, feature extraction, and imbalance handling techniques.

Authors	Core Models / Algorithms	Learning Paradigms	Feature Extraction & Engineering	Dimensionality Reduction & Filtering	Data Balancing & Imbalance Strategy
Almazroi & Ayub (2023) [1]	**RXT-J Classifier**: ResNeXt block (split cardinality convolutions) coupled with Gated Recurrent Unit (GRU) layers; benchmarked against BERT, DenseNet121, CapsNet, SVM-GS, RF, XGB.	Supervised deep sequential classification + Unsupervised representation learning + Population-based metaheuristic optimization.	**EARN Ensembler**: Combines Autoencoders (unsupervised bottleneck features) and pre-trained ResNet feature vectors via concatenation or weighted averaging; One-Hot Encoding.	Principal Component Analysis (PCA), K-Means Clustering with Silhouette score validation, and Isolation Forest Model (IFM) for anomaly score generation.	Synthetic Minority Oversampling Technique (**SMOTE**) applied during data preprocessing prior to deep feature extraction.
Ashtiani & Raahemi (2022) [2]	Analyzes 8 algorithmic classes: SVM (31 papers), Decision Trees (24), Naive Bayes / BBN (19), ANN / MLFF (15), LR (14), Ensembles (14: RF, Bagging, AdaBoost, Stacking), Metaheuristics (GP, Firefly), GMDH, KNN.	Predominantly Supervised (42/47); Unsupervised (4: SOM, GHSOM, K-means, SVD clustering, Mahalanobis distance); Semi-Supervised (1: Multi-instance / semi-supervised SVM).	Extraction of 5 financial ratio groups (leverage, liquidity, efficiency, profitability, market value) + 20 non-financial ratios (governance, auditor BIG4, restatements) + text mining (MD&A SVD).	Singular Value Decomposition (SVD), Stepwise Regression, Genetic Algorithm-based feature selection, FURIA fuzzy rule elimination.	Identified major gap: majority used raw imbalanced datasets without correction; few applied SMOTE; under-sampling completely neglected due to choice-based sampling bias.
Chen et al. (2026) [3]	Surveys DL architectures: CNNs, LSTMs, GRUs, Transformers (BERT), GNNs (GCN, ASA-GNN), Generative models (GANs, VAEs), and Hybrids (CatBoost-DNN, RDQN, Transformer-LOF-RF).	Supervised DL, Graph representation learning, Generative adversarial modeling, Reinforcement learning (Deep Q-Networks), and Semi-supervised learning.	Time-series transactional embeddings, graph node/edge topology representation (user-device-transaction rings), NLP textual analysis (sentiment, readability, Word2Vec, BERT).	PCA, Autoencoder-based latent compression, rough set theory, and automated subsampling pipelines.	SMOTE, ADASYN, Cluster-based oversampling, Generative Adversarial Networks (GANs), and Variational Autoencoders (VAEs) for realistic synthetic minority generation.
Hernandez Aros et al. (2024) [4]	Synthesizes top 10 models across 86 experimental works: RF (34 mentions), LR (32), DT (29), SVM (29), NB (19), ANN (17), KNN (14), XGBoost (13), Autoencoders (10), LSTM (7).	Supervised (56.7%), Unsupervised (18.3%), Supervised-Unsupervised combinations (15.4%), Deep Learning (2.9%), Semi-supervised (1.9%).	Transaction-level engineering (amount, timestamp, merchant category, frequency, card type) and accounting indicators (asset quality, sales growth, debt-to-equity).	PCA (primary in credit card benchmarks), Genetic Algorithms (GA) for feature selection, DBSCAN, and Isolation Forest.	Highlights SMOTE as dominant technique (e.g., in BankSim and Credit Card benchmarks); flags scarcity of cost-sensitive learning in published pipelines.
Jin & Zhang (2025) [5]	**Dynamic Stacking Ensemble**: Multi-tier model integrating base learners (LR, DT, RF, GBT, SVM, NN) mapped to a secondary meta-learner; benchmarked against SOTA baselines.	Supervised Stacking Ensemble learning with embedded cross-validation dynamic weight updating.	Transaction dynamics: time-window sliding, transaction interval, spending frequency, amount change, province single-hot coding, tag coding, user behavioral profiling.	**Recursive Feature Elimination (RFE)** with base models to iteratively prune lowest-scoring features based on importance scores.	Standardization and min-max normalization; future exploration noted for SMOTE and GANs to expand rare fraud instances.

2.3 Table 3: Datasets, Modalities, and Data Characteristics

Table 3: Comparative analysis of datasets, data modalities, sample volumes, and class distribution ratios.

Authors	Primary Datasets Investigated	Data Modalities	Scale / Sample Size	Imbalance Ratio (Fraud : Total)	Data Access, Origins & Geographic Scope
Almazroi & Ayub (2023) [1]	1. IEEE-CIS Fraud Detection 2. PaySim Synthetic Mobile Money 3. UCI European Credit Card	Structured numerical features, anonymized transaction variables, time-series intervals.	- IEEE-CIS: 590,000 records, 433 features - PaySim: 600,000 samples, 11 features - UCI: 30,000 records, 23 features	- IEEE-CIS: $\approx 3.5\%$ fraud - PaySim: Extreme minority fraud ($< 0.2\%$) - UCI: Highly imbalanced ($\approx 3.27\%$)	Public Kaggle competitions and GitHub/UCI repositories; international transaction patterns, synthetic African mobile money logs.
Ashtiani & Raahemi (2022) [2]	Stock Exchange registries, SEC AAERs, TEJ (Taiwan), Audit Analytics, COMPUSTAT, SeekingAlpha, CSRC (China), Lebanese audit software.	Structured (77%): Financial/non-financial ratios; Unstructured (23%): MD&A text (20%), CEO vocal speech/LVA audio (3%).	Majority of studies used small samples (< 260 firms); largest dataset had 15,985 statements (51 fraud vs. 15,934 non-fraud).	Severely imbalanced (fraud ratios as low as 0.003 / 0.32%); only 14 studies used artificially balanced 1:1 cohorts.	Real corporate regulatory archives from US (SEC), China (CSRC), Taiwan (TEJ), Greece (Athens SE), Lebanon, Vietnam, India, Korea.
Chen et al. (2026) [3]	European Credit Card, synthetic insurance claims, Ethereum/Bitcoin blockchain graphs, banking transaction logs, NPHIES healthcare records.	Multi-modal: Tabular transaction logs, sequential event histories, heterogeneous entity graphs, claims narratives, financial text (10-K).	Ranging from tens of thousands of tabular rows to millions of graph edges and high-frequency trading streams.	Extreme imbalance across 92/108 studies ($< 0.1\%$ to 1% typical in real-world credit and AML banking datasets).	Heavily skewed toward non-public private institutional data (85 non-public vs. 23 public); US, EU, and emerging Saudi Arabia initiatives.
Hernandez Aros et al. (2024) [4]	Top datasets: European Credit Card (15 papers), Statlog German (5), Australian Credit (4), CSMAR (3), PaySim (3), Default Taiwan (3), BankSim (2), Compustat (2), COIL 2000 (2), Bitcoin (1).	Primarily tabular numeric variables, PCA-transformed features, categorical attributes; limited textual disclosures.	From small legacy sets (690 Australian, 1,000 German) to massive benchmarks (284,807 European CC; 6.36M PaySim; 594k BankSim).	Varies from balanced credit risk benchmarks to extreme real-world transaction skew (European CC: 492 fraud / 284,807 total $= 0.172\%$).	Real data dominates ($>$ 93% of studies); stock exchanges of China, US, Taiwan, Canada, Tehran; public UCI/Kaggle repositories.
Jin & Zhang (2025) [5]	Multi-channel financial database: 10 financial institutions + Kaggle/UCI public data + 5 commercial market data providers; cross-tested on IEEE-CIS, PaySim, Alipay SEC.	Structured numerical transaction variables, multi-dimensional time windows, categorical tags, trading place indicators.	**Over 1,000,000 primary records**: - Institutional: 600,000 - Kaggle/UCI: 300,000 - Market providers: 100,000 - Cross-domain test: 4.2M records	Mixed real-world operational distributions containing legitimate transfers, credit card theft, and synthetic transaction fraud patterns.	Hybrid real institutional data from Chinese commercial banks and fintech platforms combined with standardized open-source benchmarks.

2.4 Table 4: Quantitative Performance, Evaluation Metrics, and Operational Trade-Offs

Table 4: Evaluation metrics, peak empirical performance, operational latency, computational overhead, and statistical validation.

Authors	Primary Evaluation Metrics	Reported Peak Performance	Comparative Benchmark Outcomes	Operational Latency & Computing Overhead	Statistical Validation Rigor
Almazroi & Ayub (2023) [1]	Accuracy, Precision, Recall, F1-Score, Matthews Correlation Coefficient (MCC), ROC Loss, AUC-ROC.	**IEEE-CIS**: Acc 97.9%, Rec 99.3%, Prec 97.7%, F1 0.987, MCC 0.984, AUC 0.991. **European**: Acc 98.1%, F1 0.989, Rec 99.5%, AUC 0.981.	Consistently outperformed baselines by 10%–18%: BERT (Transf.) Acc 91.2%, ResNet 92.4%, CapsNet 91.2%, DenseNet121 89.1%, SVM-GS 85.9%, XGB 14.0%.	Demonstrated the shortest average execution time across varying dataset scales; optimized for real-time inference on GPU.	Silhouette score cluster validation; hypothesis testing on feature subsets pre- and post-SMOTE.
Ashtiani & Raahemi (2022) [2]	Classification Accuracy (%), Sensitivity (Recall), Specificity, Area Under ROC (AUC).	Deep dense Multi-Layer Perceptron (MLP) achieved peak accuracy of **93.7%** (on Greek data [7]); Logistic Regression achieved **93.3%** [6].	Single models slightly outperformed hybrids: MLP (93.7%) > Boosting (91.8%) > SOM+K-Means (91.0%) > CART (90.8%) > RF (87.5%) > Stacking (83.0%).	Not quantitatively benchmarked across surveyed studies; highlighted that deep learning and GMDH require heavy computational resources.	Meta-analysis of historical accuracy; noted severe lack of cross-validation standardization across literature.
Chen et al. (2026) [3]	Shift from pure Accuracy to Precision, Recall, F1, AUC-ROC, **AUC-PR**, False Positive Rate (FPR), False Negative Rate (FNR), and **Economic Cost of FPs/FNs**.	Surveys top performers across sectors: Hybrid CatBoost-DNN and Transformer-LOF-RF achieve SOTA F1 and AUC-PR on imbalanced streams.	GNNs (GCN, ASA-GNN) outperform classical RF and XGBoost on relational fraud rings; Transformers outperform LSTMs on long-range transaction logs.	Analyzes trade-offs: Credit card fraud requires < 10 ms latency at POS; banking and insurance prioritize auditability and compliance over ultra-low latency.	Systematic risk-of-bias evaluation across selection, performance, dataset, and reproducibility dimensions.
Hernandez Aros et al. (2024) [4]	Supervised: Accuracy, Precision, Recall/Sensitivity, Specificity, F1, AUC, Type I/II error; Unsupervised: Silhouette, Rand Index (RI), Davies-Bouldin (DB), Cophenetic Correlation (CPC).	Random Forest and Ensemble XGBoost models report peak accuracies between **90%–99%** on European Credit Card and BankSim benchmarks.	Ensemble tree models (RF, XGBoost) consistently outperform single DT and SVM in accuracy and resilience to noise on tabular benchmarks.	Reviews training time and computational resource burdens of deep neural networks vs. tree ensembles across 86 experimental setups.	Bibliometric co-citation link strength mapping; quality assessment scoring via Roehrs et al. (2017) protocol.
Jin & Zhang (2025) [5]	Accuracy, Recall, F1-Score, AUC-ROC, Precision-Recall (P-R) curves, Single Prediction Delay (ms), Training Time (hours), Memory (GB).	**Primary test**: Acc 95.0%, Rec 93.0%, F1 94.0%, AUC 0.896. At FPR=0.1, achieves 92.5% TPR (5%–12% higher fraud capture than single models).	Stacking significantly outperformed base models: Stacking (0.95) > GBT (0.92) > RF (0.90) > DT (0.87) > LR (0.85); cross-domain beat TabTransformer-FT (0.928) and GNN-Fraud (0.931).	**Inference delay**: 5.8 ms (Stacking) vs. 0.1 ms (LR), 0.3 ms (DT), 1.2 ms (RF), 2.5 ms (GBT); training time 8.0 h; memory footprint 7.5 GB.	Paired t-test confirmed statistical significance ($p < 0.001$, t-values 3.75 to 8.67) across all metric comparisons.

2.5 Table 5: Regulatory Governance, Explainability (XAI), and Future Research Directions

Table 5: Governance, regulatory privacy compliance, XAI techniques, limitations, and future research roadmaps.

Authors	Regulatory Compliance & Data Privacy	Explainability & Interpretability (XAI)	Identified Limitations & Biases	Proposed Solutions & Practical Guidance	Emerging Trends & Future Directions
Almazroi & Ayub (2023) [1]	Addresses wireless communication security, data availability, and cyber defense against transaction interference.	Features filtered via PCA and Isolation Forest anomaly scores; model internals remain largely black-box deep representations.	Risk of synthetic sample overfitting from SMOTE; PCA linear projections may discard non-linear patterns; generalizability anchored to IEEE-CIS properties.	Provides hyperparameter convergence boundaries through the Jaya Algorithm; advocates automated ensembling for banking networks.	Incorporating geographic spatial coordinates and dynamic temporal transaction timing analysis as streaming data becomes available.
Ashtiani & Raahemi (2022) [2]	Highlights corporate disclosure regulations (SEC AAERs, CSRC filings, stock exchange oversight mandates).	Notes that Logistic Regression remains heavily favored by accounting researchers precisely due to statistical coefficient interpretability.	Small sample sizes ($<$ 260 records); language bias (English-only); high search bias toward accounting journals; scarce unsupervised validation.	Recommends integrating qualitative auditor notes, board structure, and financial social media (SeekingAlpha) into unified feature vectors.	Application of unsupervised anomaly detection (Mahalanobis distance), bio-inspired metaheuristics (AIS, GA), and NLP embeddings (BERT, Doc2Vec).
Chen et al. (2026) [3]	Core pillar of **DLSG**: GDPR "right to explanation", CCPA data minimization, HIPAA/NPHIES in healthcare; federated learning and blockchain smart contracts.	Mandatory XAI integration: SHAP values, LIME, counterfactual explanations (NICE), and feature attribution dashboards for regulatory auditing.	Exclusion of pre-2019 literature; heavy reliance on synthetic benchmarks; lack of real-world proprietary banking models in published literature.	Operationalize DLSG via modular APIs, continuous concept drift monitoring pipelines, and privacy-preserving federated training architectures.	Quantum Graph Neural Networks (QGNNs), multimodal foundation models, self-learning meta-optimizers, and industry–regulatory benchmark leaderboards.
Hernandez Aros et al. (2024) [4]	Reviews anti-money laundering (FATF standards in Saudi Arabia, Taiwan, Mexico) and tax audit frameworks (Serbia, Belgium, Armenia).	Emphasizes need for explainability to prevent algorithmic discrimination and bias in loan approvals and insurance claims.	Use of obsolete datasets (some dating back to 1994, e.g., German/Australian credit); severe geographic concentration in China/India/Saudi Arabia; 18 review papers limited raw data.	Call for creating updated, freely accessible real-world transactional benchmarks reflecting modern cyber-enabled financial fraud schemes.	Expanding research into reinforcement learning, deep unsupervised anomaly clustering, and non-banking fraud (embezzlement, smurfing, document forgery).
Jin & Zhang (2025) [5]	User data anonymization in compliance with financial privacy regulations across 10 partner banking institutions.	Utilizes feature importance scores derived from base learners to enhance partial transparency; acknowledges black-box nature of deep stacking layer.	Higher computational latency (5.8 ms per transaction) and significant GPU/RAM memory footprint (7.5 GB), presenting production deployment bottlenecks.	Recommends balancing real-time latency requirements against detection accuracy thresholds depending on transaction risk tiers.	Integration of online incremental learning to combat concept drift; multimodal fusion incorporating text, audio, and visual KYC checks; GAN-based synthetic data generation.

3 Synthesis of Comparative Insights and Discussion

A comprehensive examination of the five studies reveals several overarching paradigms, trade-offs, and shifts in the financial fraud detection landscape:

3.1 Architectural Shift: From Single Classifiers to Hybrid and Deep Sequential Models

The early literature reviewed by Ashtiani & Raahemi (2022) and Hernandez Aros et al. (2024) demonstrates an overwhelming reliance on single supervised algorithms, particularly Support Vector Machines, Logistic Regression, and Decision Trees. While computationally accessible and statistically interpretable, single models exhibit high false positive rates and fail to capture non-linear transaction relationships.

In contrast, the recent empirical contributions of Almazroi & Ayub (2023) and Jin & Zhang (2025) confirm that advanced ensembling and hybrid deep architectures are necessary to achieve commercial-grade detection ($> 95\%$ accuracy and recall). Almazroi & Ayub demonstrate that combining spatial convolution (ResNeXt) with sequential memory (GRU) resolves temporal transaction shifts, while Jin & Zhang prove that heterogeneous Stacking with dynamic weighting captures multi-level fraud patterns that individual algorithms miss.

3.2 The Imbalance–Overfitting Dilemma

All five studies identify extreme class imbalance as the primary obstacle to reliable fraud detection. While SMOTE remains the de facto standard (utilized in Almazroi & Ayub, 2023, and extensively surveyed by Chen et al., 2026), several authors caution against synthetic sample overfitting. Chen et al. (2026) highlight the emerging adoption of Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) as superior alternatives capable of generating realistic minority class transactions while preserving the underlying distribution topology.

3.3 Operational Feasibility: The Latency vs. Accuracy Trade-Off

A crucial insight highlighted by Jin & Zhang (2025) and conceptualized in Chen et al.'s (2026) DLSG framework is the operational trade-off between algorithmic sophistication and decision latency:

- In point-of-sale (POS) credit card verification, decision engines must return inferences within < 10 milliseconds. While Jin & Zhang's Stacking model achieved 95% accuracy, its 5.8 ms delay and 7.5 GB memory footprint approach operational limits for ultra-high-frequency streams.
- Conversely, in corporate financial statement fraud (Ashtiani & Raahemi, 2022) and tax evasion monitoring (Hernandez Aros et al., 2024), batch processing permits higher computational overhead, making deep multimodal and graph models viable.

3.4 The Regulatory Imperative and Explainable AI (XAI)

As formulated in Chen et al. (2026), modern financial fraud detection cannot operate purely as an optimization exercise. Strict compliance with GDPR, CCPA, and sector mandates requires models to provide actionable explanations for flagged transactions. While traditional models like Logistic Regression remain favored in accounting for their transparent coefficients (Ashtiani & Raahemi, 2022), modern systems must bridge deep learning with post-hoc interpretability frameworks like SHAP, LIME, and counterfactual reasoning.

4 Conclusion

This comparative analysis illustrates that the frontier of financial fraud detection lies in the convergence of **deep sequential and graph architectures**, **dynamic ensemble strategies**, and **governance-aligned deployment pipelines**. While models such as RXT-J and Dynamic Stacking deliver unprecedented empirical accuracy, future research must prioritize lightweight online learning to mitigate latency, privacy-preserving federated learning across banking consortia, and rigorous XAI integration to satisfy legal compliance.

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